팹리스 실무형 일경험 프로그램

SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습

9일차

02 Sensing Zone과 HPC Zone 연동 구현

목차

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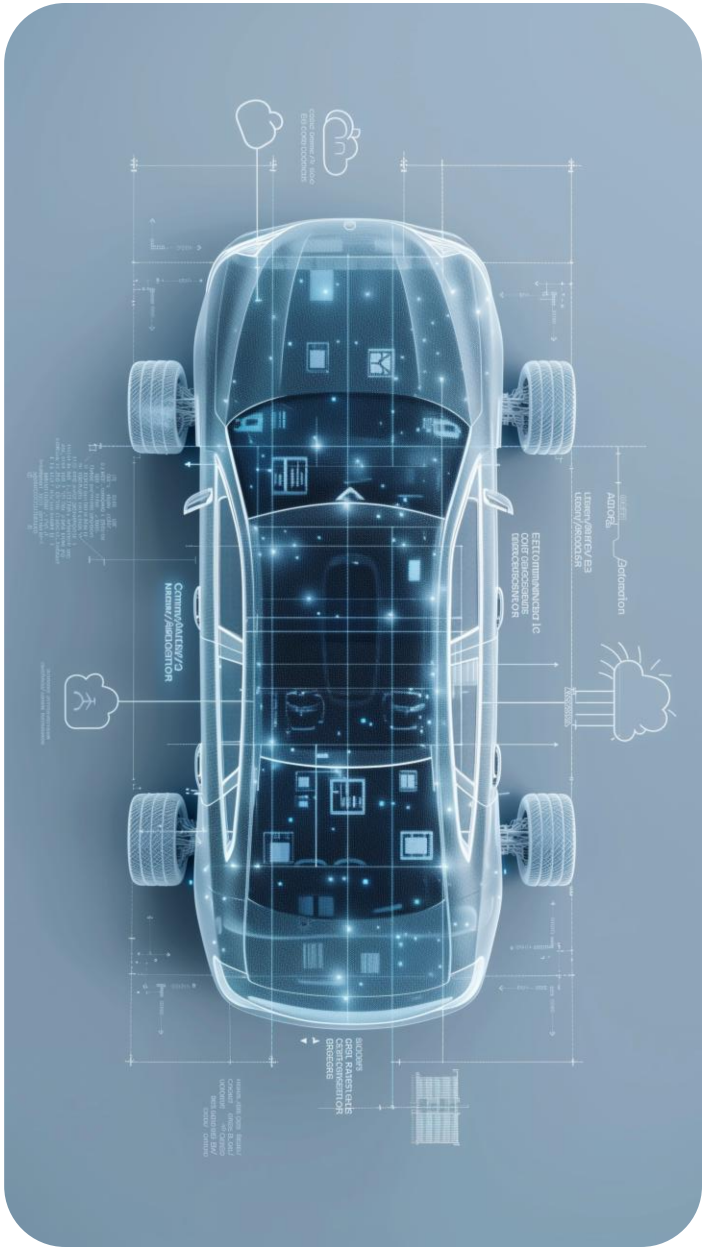
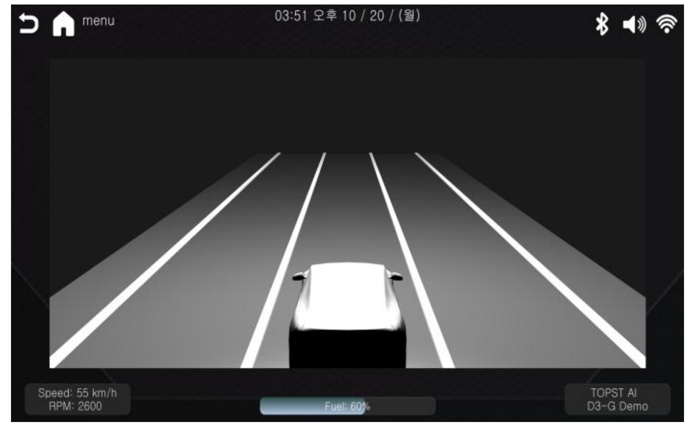


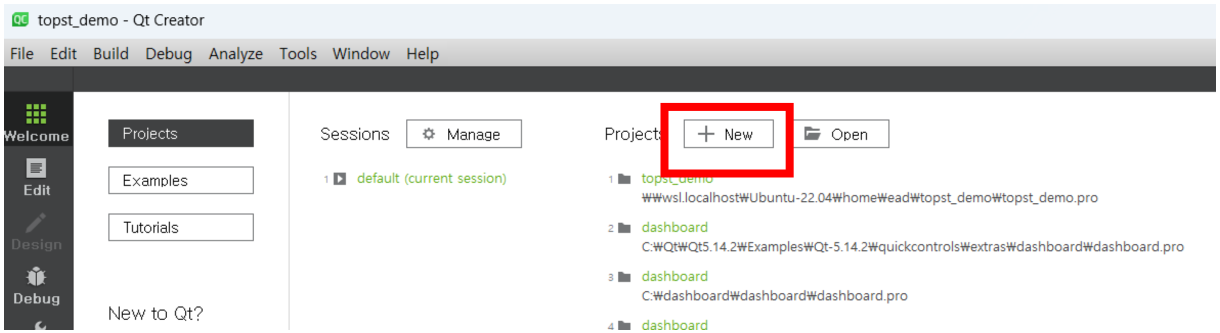
Panel App 동작 구조

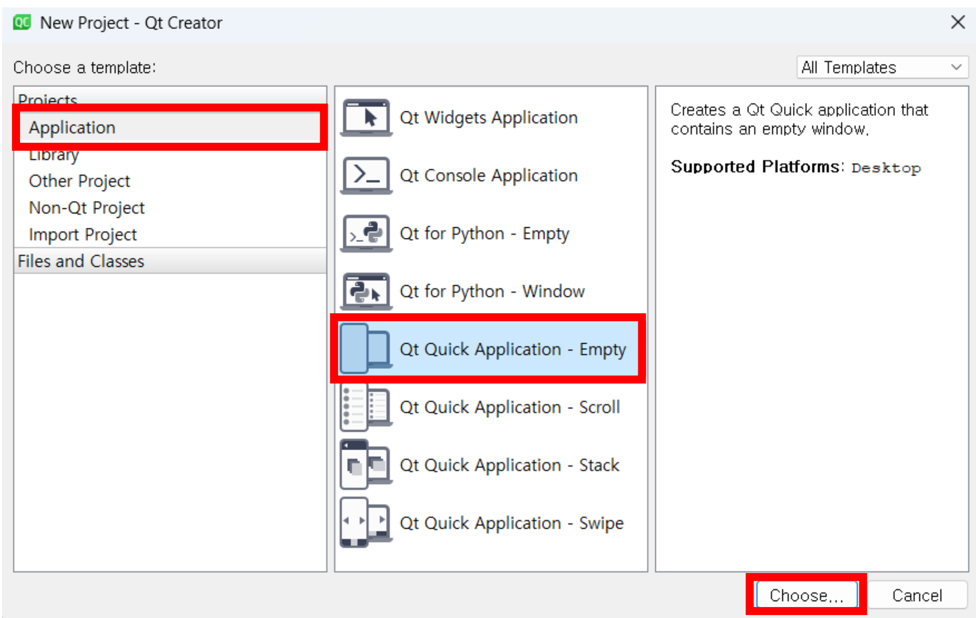


- Yolov8 모델 추론
- 추론된 바운딩 박스의 좌표 값으로 중심 좌표 계산
- 중심 좌표 위치에 따라 특정 위치 매핑 -> 차량 모델 출력

Sensing Zone + HPC Zone

Panel App 구현

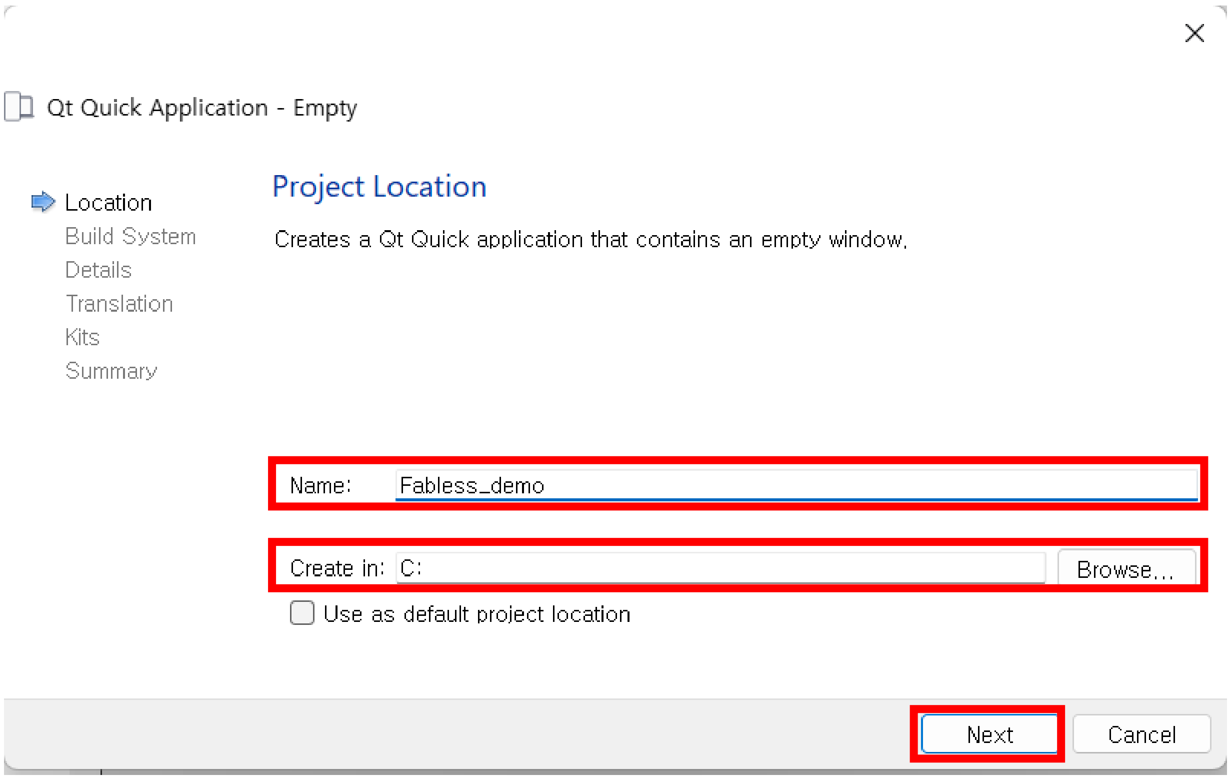




- QT creator 실행
- New 탭 선택
- Application -> Qt Quick Application 선택
- Choose 클릭

Sensing Zone + HPC Zone

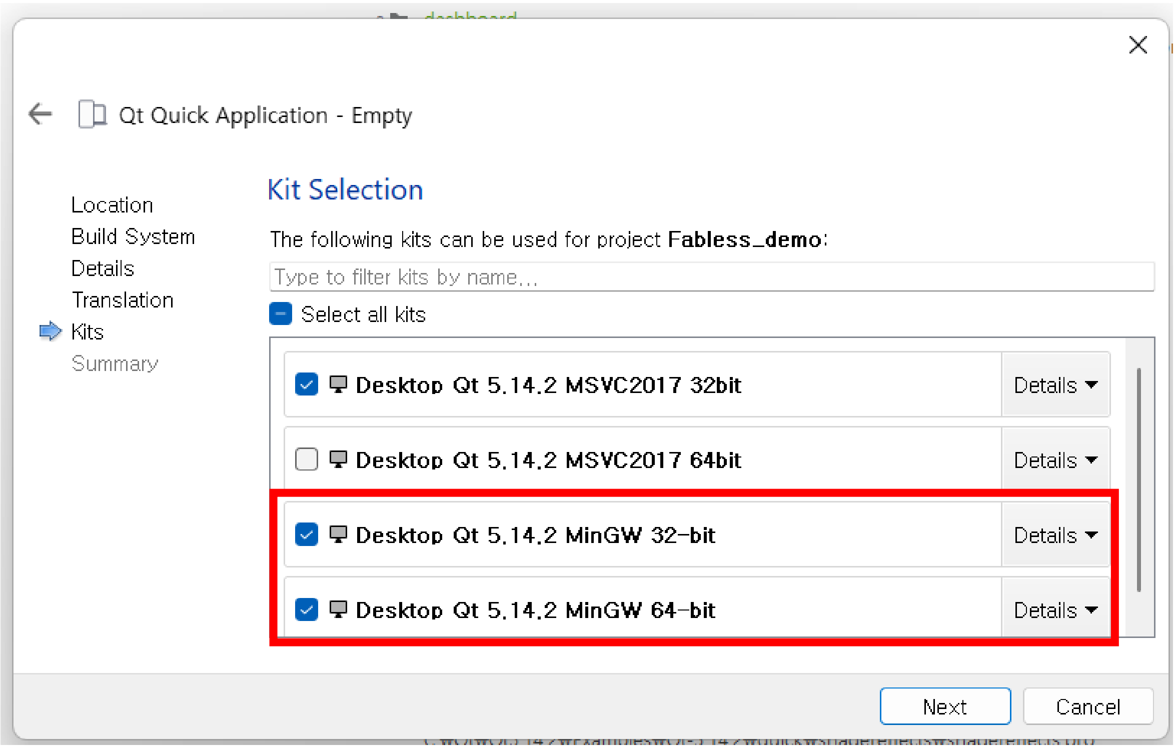
Panel App 구현



- Name : Fabless_demo 수정
- Create in : 로컬 디스크 C
- Next -> Next -> Next
- ->Next

Sensing Zone + HPC Zone

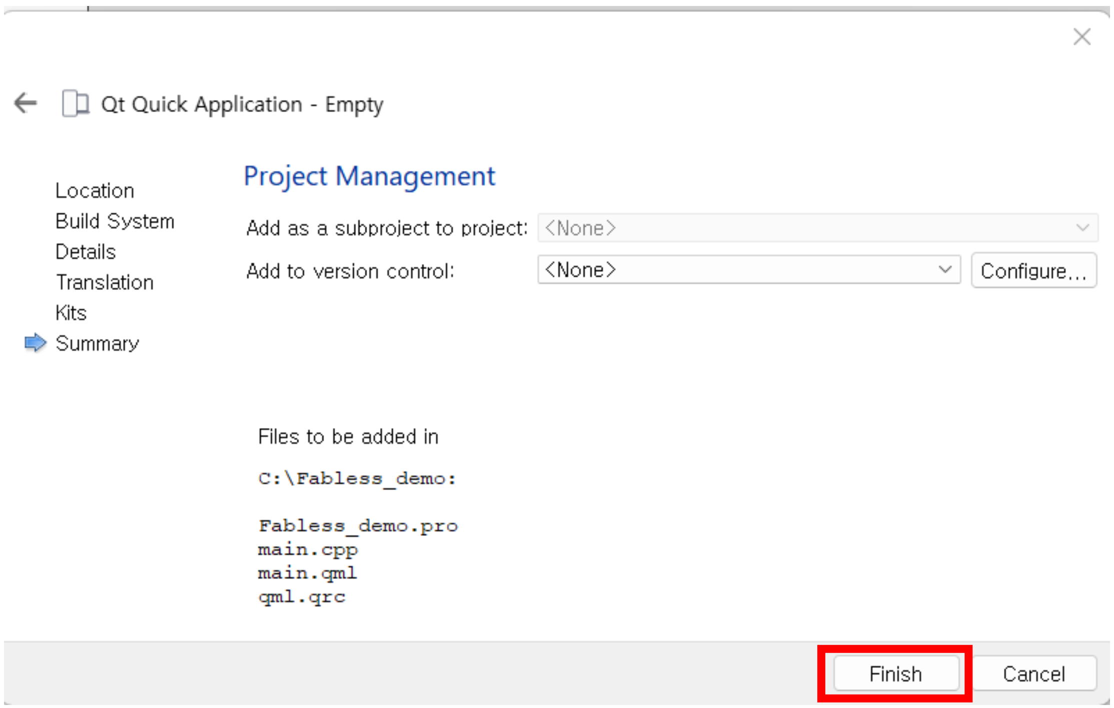
Panel App 구현



- Desktop QT 5.14.2 MinGW 32-bit / 64-bit 활성화
- Next 클릭

Telechips

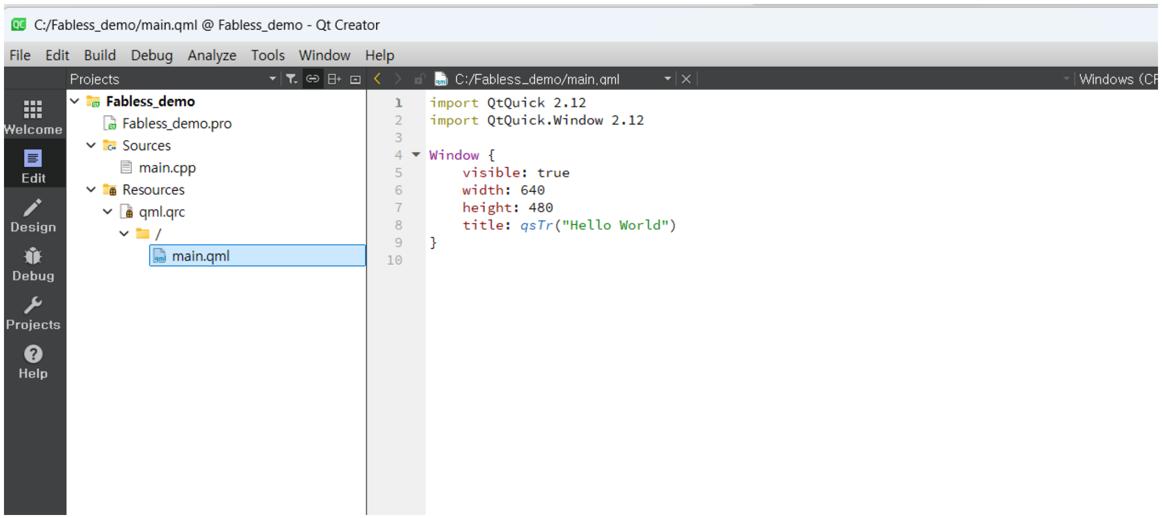
Panel App 구현



- Finish 클릭

Sensing Zone + HPC Zone

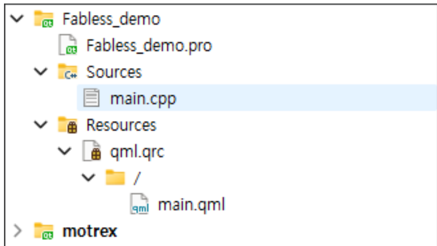
Panel App 구현



- 해당 화면 확인

Telechips

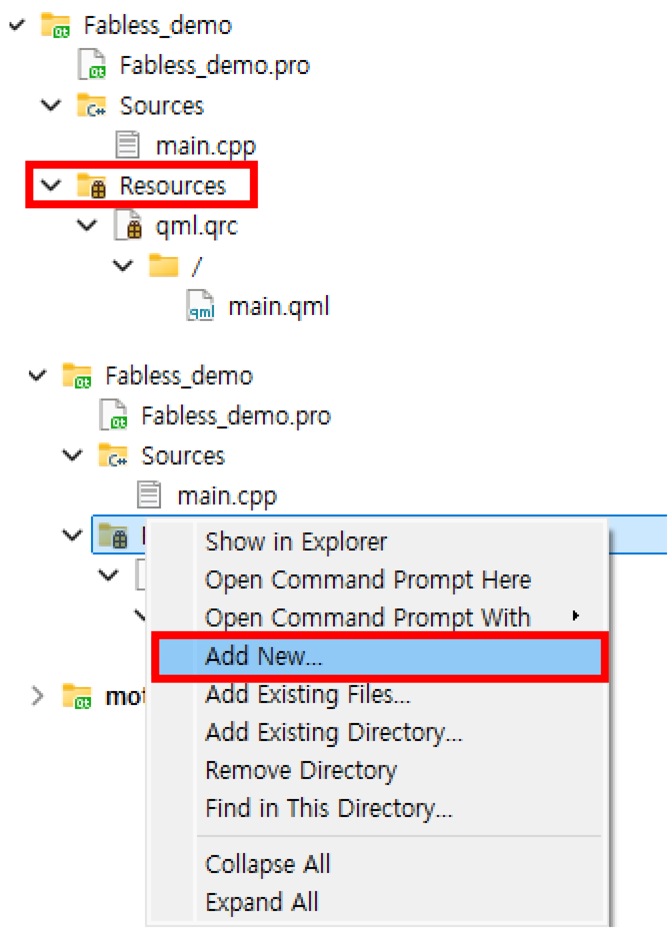
Panel App 구현




```
QT += quick

CONFIG += c++11

# The following define makes your compiler emit warnings if you use
# any Qt feature that has been marked deprecated (the exact warnings
# depend on your compiler). Refer to the documentation for the
# deprecated API to know how to port your code away from it.
DEFINES += QT_DEPRECATED_WARNINGS

# You can also make your code fail to compile if it uses deprecated APIs.
# In order to do so, uncomment the following line.
# You can also select to disable deprecated APIs only up to a certain version of
#DEFINES += QT_DISABLE_DEPRECATED_BEFORE=0x060000    # disables all the APIs depr

SOURCES += \
        main.cpp

RESOURCES += qml.qrc \
    icons.qrc

# Additional import path used to resolve QML modules in Qt Creator's code model
QML_IMPORT_PATH =

# Additional import path used to resolve QML modules just for Qt Quick Designer
QML_DESIGNER_IMPORT_PATH =


# Default rules for deployment.
qnx: target.path = /tmp/$${TARGET}/bin
else: unix:!android: target.path = /opt/$${TARGET}/bin
!isEmpty(target.path): INSTALLS += target

DISTFILES += \
    config.json
```

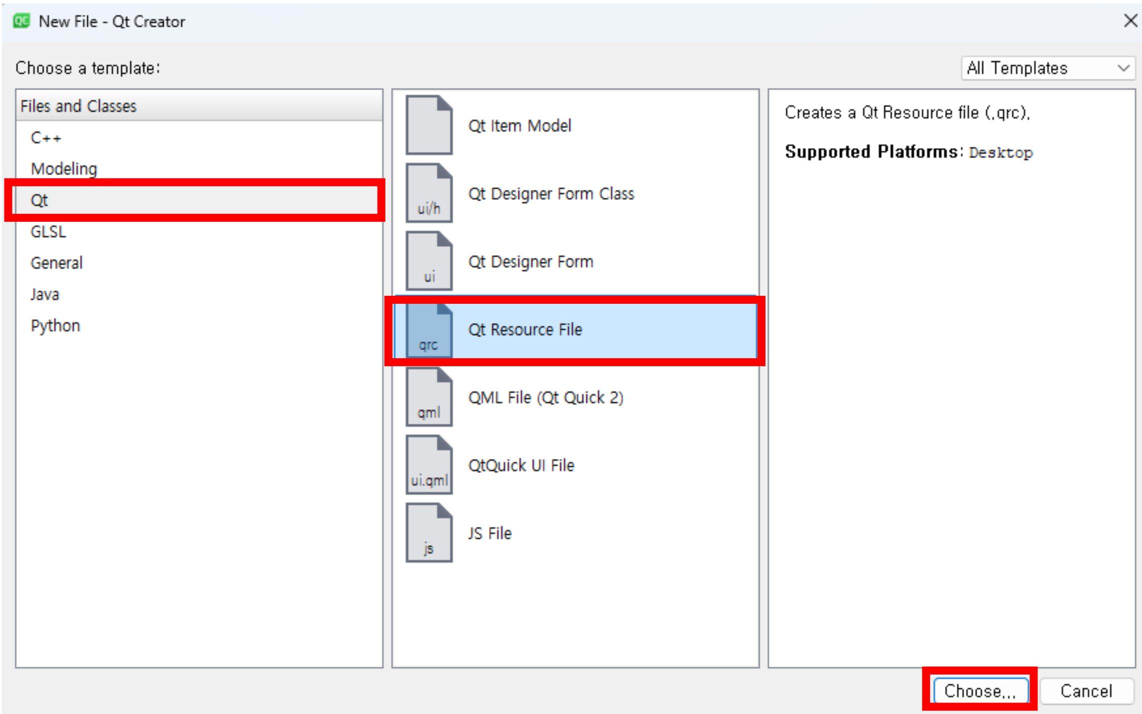
- Fabless_demo.pro 선택
- RESOURCES에 `icons.qrc` 추가

Panel App 구현



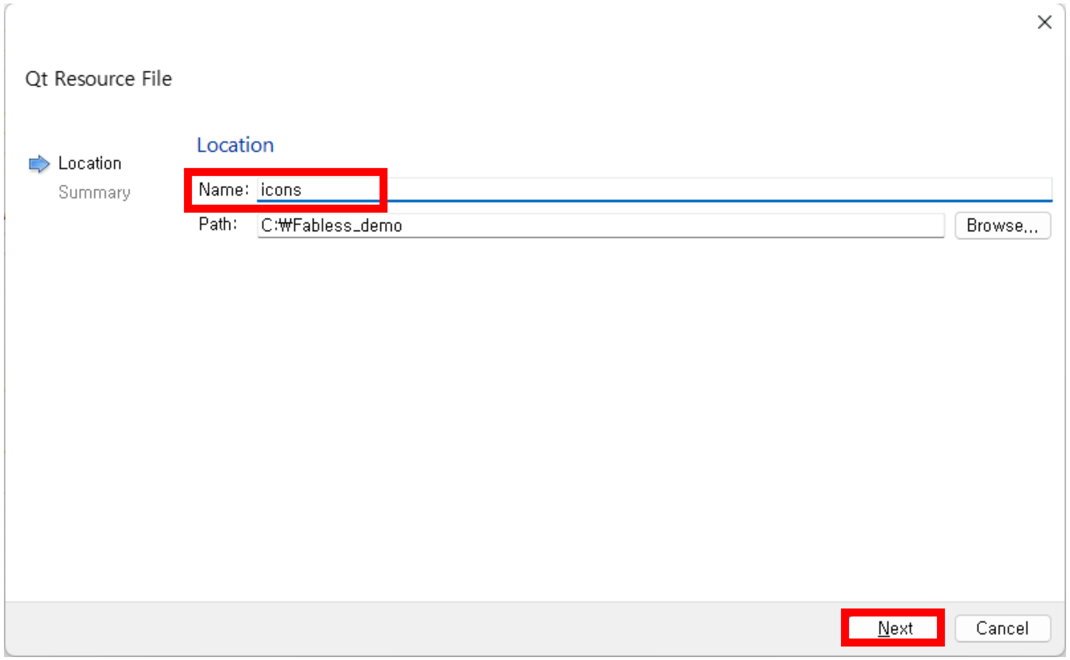
- Resources 우클릭
- Add New... 선택

Panel App 구현



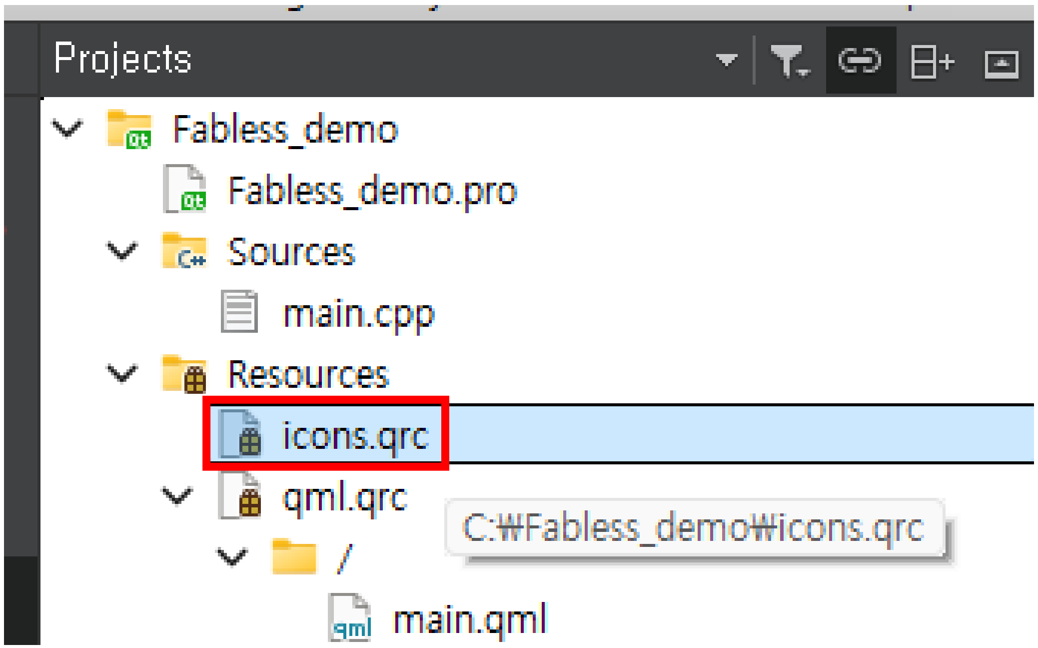
- Qt -> Qt Resources File 선택
- Choose 클릭

Panel App 구현



- Name – icons 입력
- Next -> Finish

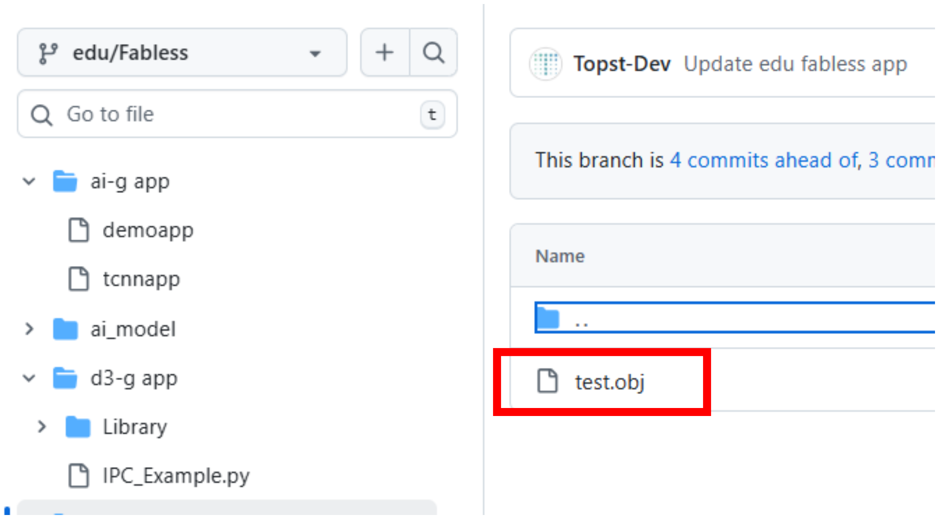
Panel App 구현

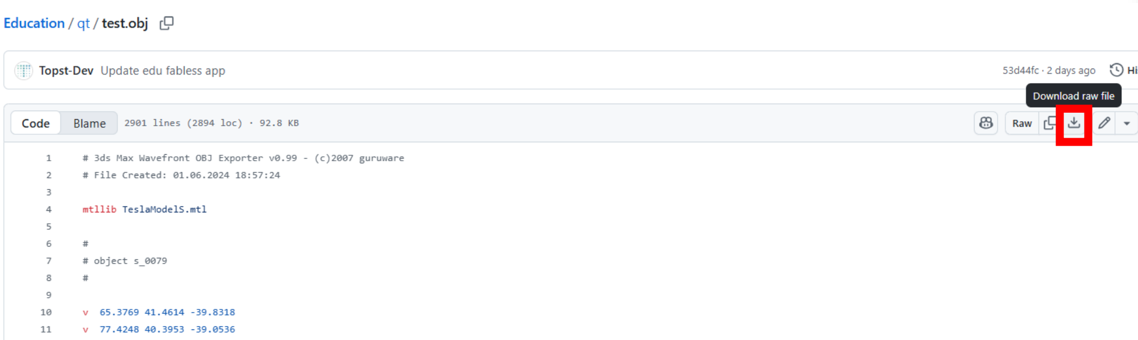


- icons.qrc 확인

Sensing Zone + HPC Zone

Panel App 구현

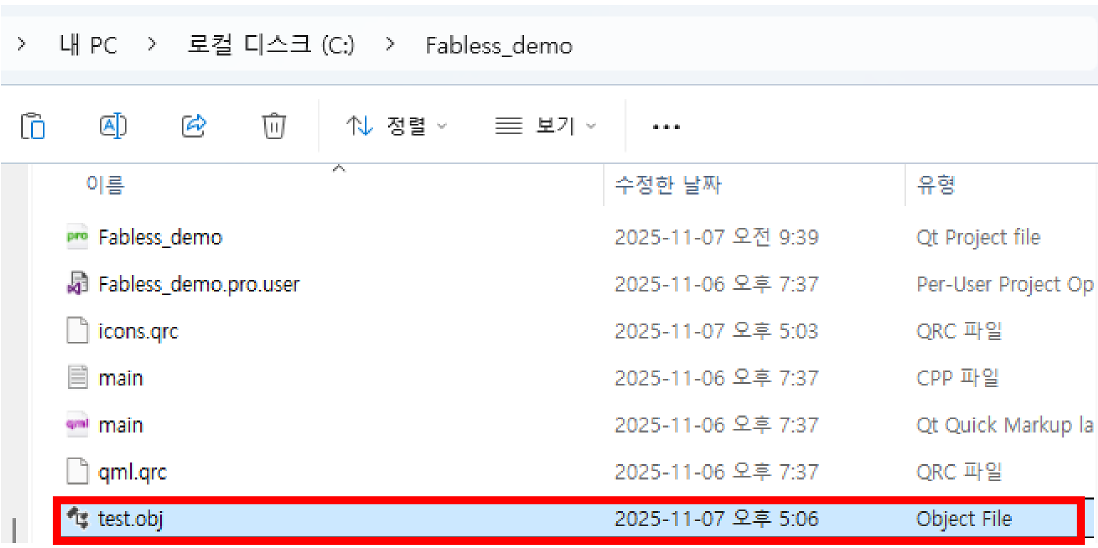




- [Education/qt at edu/Fabless · topst-development/Education]
- 해당 링크 접속
- test.obj 선택
- 다운로드 버튼 클릭

Telechips

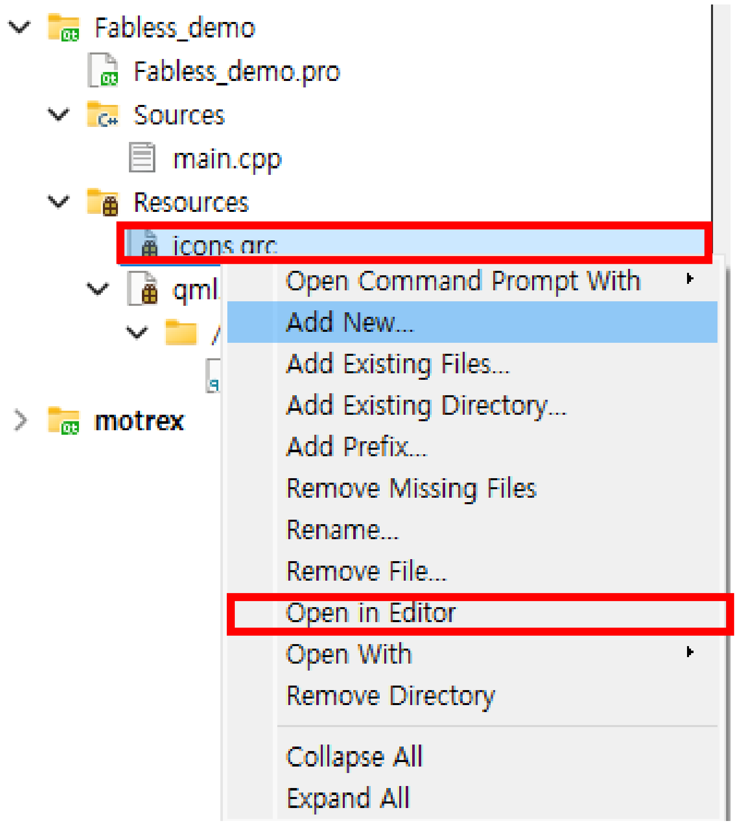
Panel App 구현



- 해당 .obj 파일을 현재 작업 중인 qt 디렉터리로 복사

Telechips

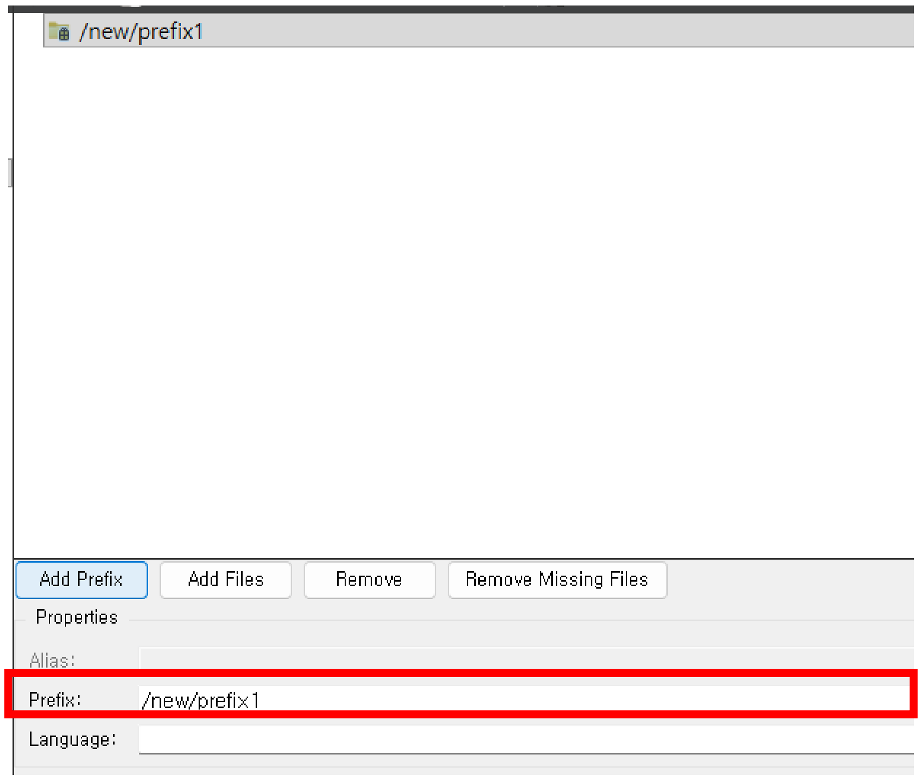
Panel App 구현



- icons.qrc 오른쪽 클릭
- Open in Editor클릭

Sensing Zone + HPC Zone

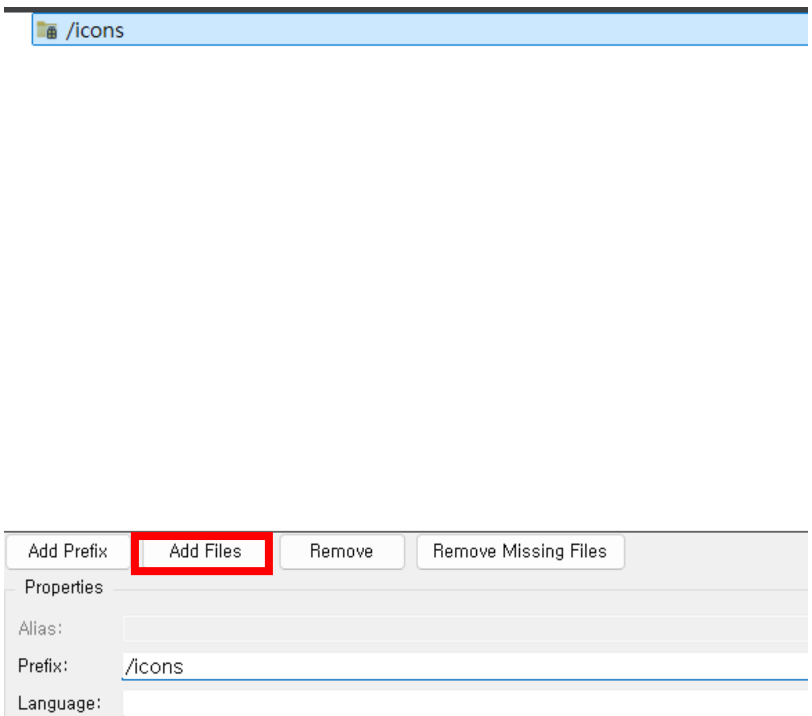
Panel App 구현



- Add Prefix 선택
- Prefix 를 /new/prefix1 -> /icons 로 수정

Telechips

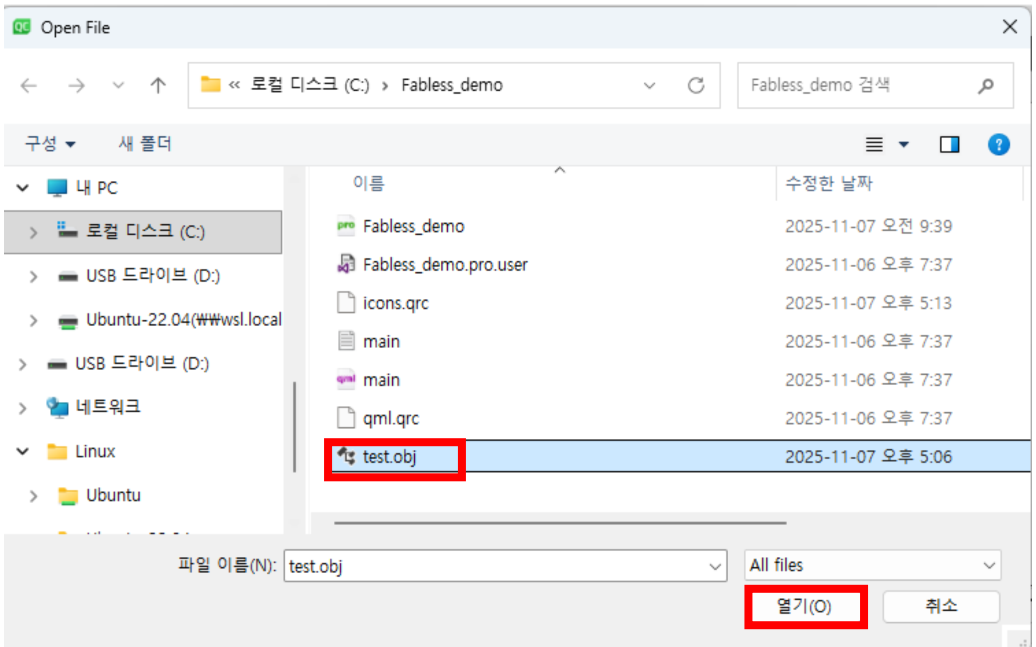
Panel App 구현



- Add Files 선택

Sensing Zone + HPC Zone

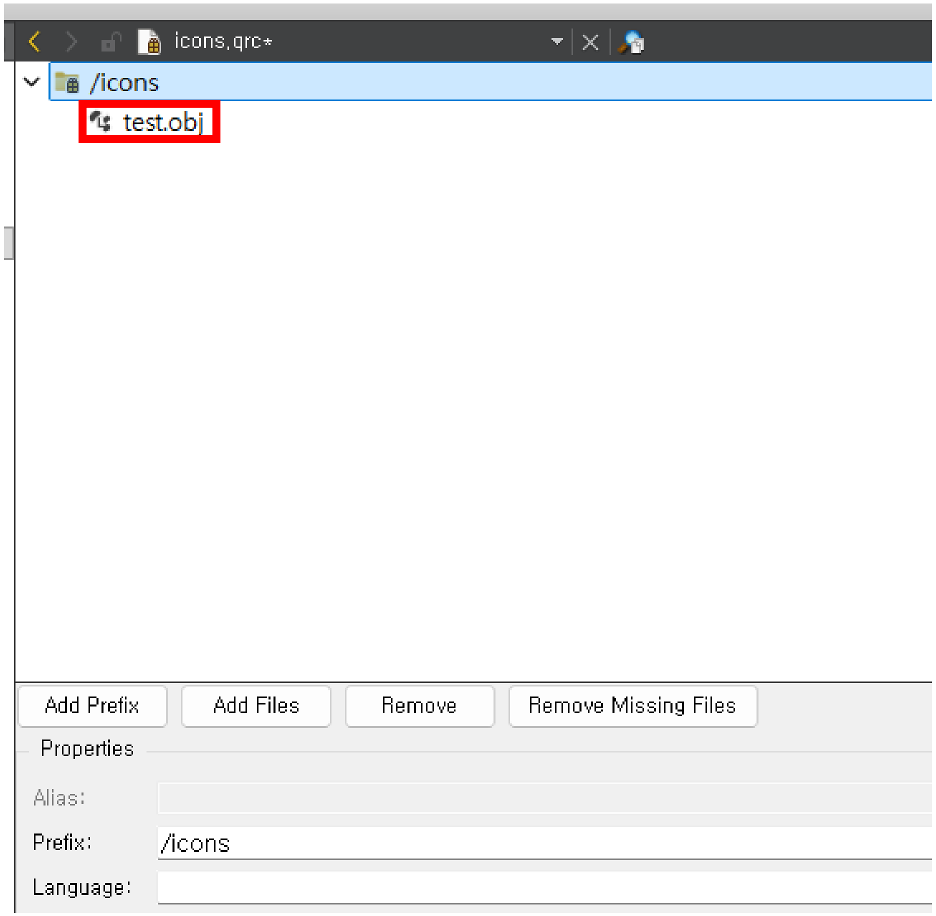
Panel App 구현



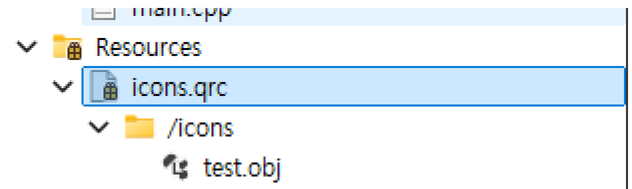
- test.obj 선택
- 열기 클릭

Telechips

Panel App 구현



- 해당 화면에서 ctrl+s 입력
- 아래와 같이 생성되는지 확인



Panel App 구현

- Sources/main.cpp 코드 수정

```
#include <QGuiApplication>
#include <QQmlApplicationEngine>
#include <QTcpSocket>
#include <QTimer>
#include <QQmlContext>
#include <QDebug>
#include <QtGui/QFont>
#include <QtGui/QFontDatabase>
#include <QFile>
#include <QJsonDocument>
#include <QJsonObject>
#include <QCommandLineParser>
#include <QCommandLineOption>
#include <QSurfaceFormat>
#include <QGuiApplication>
#include <QQmlApplicationEngine>
#include <QQuickView>
```

```
class TcpClient : public QObject {
    Q_OBJECT
public:
    TcpClient() {
        connect(&socket, &QTcpSocket::connected, this, [](){
            qDebug() << "✅ 서버에 연결 성공!";
        });
        connect(&socket, &QTcpSocket::disconnected, this, [](){
            qDebug() << "⚠️ 서버와 연결이 끊어졌습니다.";
        });
        connect(&socket, &QTcpSocket::readyRead, this,
&TcpClient::checkForData);
    }

    Q_INVOKABLE void connectTo(const QString& ip, int port) {
        if (socket.isOpen() || socket.state() ==
QAbstractSocket::ConnectingState)
            socket.disconnectFromHost();
        qDebug() << "➡️  서버 접속 시도:" << ip << port;
        socket.connectToHost(ip, port);
    }
```

Panel App 구현

- Sources/main.cpp 코드 수정

```
    void connectFromConfig() {
        QFile file(":/icons/config.json");
// qrc 기본값
        if (file.open(QIODevice::ReadOnly)) {
            auto json =
QJsonDocument::fromJson(file.readAll()).object
();
            const QString ip =
json.value("server_ip").toString();
            const int port    =
json.value("server_port").toInt();
            connectTo(ip, port);
        } else {
            qWarning() << "config.json(qrc) 열
기 실패";
        }
    }

signals:
    void showCarsForLane(int lane);   // lane
번호 전달
```

```
private slots:
    void checkForData() {
        if (socket.bytesAvailable() > 0) {
            QByteArray data = socket.readAll();

            if (data.contains("object_detected_lane_1")) emit
showCarsForLane(1);
                    else if (data.contains("object_detected_lane_2"))
emit showCarsForLane(2);
                    else if (data.contains("object_detected_lane_3"))
emit showCarsForLane(3);
                    else if (data.contains("object_detected_lane_4"))
emit showCarsForLane(4);
                    else if (data.contains("object_detected_lane_5"))
emit showCarsForLane(5);
        }
    }

private:
    QTcpSocket socket;
    QTimer timer;
};
```


Sensing Zone + HPC Zone

Panel App 구현

- Sources/main.cpp 코드 수정

```
int main(int argc, char *argv[])
{
    QGuiApplication app(argc, argv);

    //qt 검증을 위해 eglfs 옵션은 잠시 주석 처리
    //QSurfaceFormat format;
    //format.setRenderableType(QSurfaceFormat::OpenGLES);
    //format.setVersion(2, 0);
    //QSurfaceFormat::setDefaultFormat(format);

    // --- 커맨드라인 파서 설정 ---
    QCommandLineParser parser;
    parser.setApplicationDescription("Lane telemetry
client");
    parser.addHelpOption();
    QCommandLineOption ipOpt({"i","ip"}, "Server IP
address", "ip");
    QCommandLineOption portOpt({"p","port"}, "Server port",
"port");
    parser.addOption(ipOpt);
    parser.addOption(portOpt);
    parser.process(app);

    QQmlApplicationEngine engine;

QFontDatabase::addApplicationFont(":/fonts/DejaVuSans.ttf");
    app.setFont(QFont("DejaVu Sans"));
```

```
    TcpClient client;
    engine.rootContext()-
>setContextProperty("tcpClient", &client);

    // 인자 우선, 없으면 qrc 설정 사용
    const QString ipArg = parser.value(ipOpt);
    bool ok = false;
    int portArg = parser.value(portOpt).toInt(&ok);
    if (!ipArg.isEmpty() && ok && portArg > 0) {
        client.connectTo(ipArg, portArg);
    } else {
        client.connectFromConfig();
    }

    engine.load(QUrl(QStringLiteral("qrc:/main.qml")));
    return app.exec();
}

#include "main.moc"
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
import QtQuick 2.12
import QtQuick.Controls 2.12
import QtQuick.Scene3D 2.12
import Qt3D.Core 2.12
import Qt3D.Render 2.12
import Qt3D.Input 2.12
import Qt3D.Extras 2.12

ApplicationWindow {
    id: win
    width: 800
    height: 400
    visible: true
    title: "Lane 3D Demo"

    property real yaw: 0
    property real pitch: 10
    property real zoom: 30
    property var laneX: [-7.5, -2.5, 2.5, 7.5]
```

```
function updateCameraPosition() {
        let radYaw = yaw * Math.PI / 180
        let radPitch = pitch * Math.PI / 180
        let x = zoom * Math.sin(radYaw) *
Math.cos(radPitch)
        let y = zoom * Math.sin(radPitch)
        let z = zoom * Math.cos(radYaw) *
Math.cos(radPitch)
        camera.position = Qt.vector3d(x, y, z)
    }
    property var lastSeenMs: ({1:0, 2:0, 3:0, 4:0, 5:0})
    // C++ setContextProperty("tcpClient", &client)
    Connections {
        target: tcpClient
        function onShowCarsForLane(lane) {
            var now = Date.now()
            lastSeenMs[lane] = now
            if (lane === 1) carEnt1.enabled = true
            if (lane === 2) carEnt2.enabled = true
            if (lane === 3) carEnt3.enabled = true
            if (lane === 4) carEnt4.enabled = true
            if (lane === 5) carEnt5.enabled = true
        }
    }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
Item {
       anchors.fill: parent

       Scene3D {
           anchors.fill: parent
           aspects: ["input", "logic", "render"]

           Entity {
               id: rootEntity

               Camera {
                   id: camera
                   position: Qt.vector3d(0, 20, 40)
                   viewCenter: Qt.vector3d(0, 0, 0)
                   upVector: Qt.vector3d(0, 1, 0)
               }
```

```
                components: [
                    RenderSettings {
                        activeFrameGraph:
ForwardRenderer {
                            camera: camera
                            clearColor: "#1a1a1a"
                        }
                    },
                    InputSettings {}
                ]

                // 조명
                Entity {
                    components: [
                        PointLight { color: "white";
intensity: 3; constantAttenuation: 1.0 },
                        Transform { translation:
Qt.vector3d(0, 10, 10) }
                    ]
                }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
                // 도로
                Entity {
                    components: [
                        PlaneMesh { width: 20; height: 80 },
                        PhongMaterial { diffuse: "#303030"; shininess: 10; specular: "#555555" },
                        Transform { rotation: fromAxisAndAngle(Qt.vector3d(1, 0, 0), 0); translation:
Qt.vector3d(0, -1.0, 0) }
                    ]
                }

                // 차선 4개
                Entity { components: [ CuboidMesh { xExtent: 0.3; yExtent: 0.2; zExtent: 60 },
PhongMaterial { diffuse: "white" }, Transform { translation: Qt.vector3d(-7.5, -1.0, -10) } ] }
                Entity { components: [ CuboidMesh { xExtent: 0.3; yExtent: 0.2; zExtent: 60 },
PhongMaterial { diffuse: "white" }, Transform { translation: Qt.vector3d(-2.5, -1.0, -10) } ] }
                Entity { components: [ CuboidMesh { xExtent: 0.3; yExtent: 0.2; zExtent: 60 },
PhongMaterial { diffuse: "white" }, Transform { translation: Qt.vector3d( 2.5, -1.0, -10) } ] }
                Entity { components: [ CuboidMesh { xExtent: 0.3; yExtent: 0.2; zExtent: 60 },
PhongMaterial { diffuse: "white" }, Transform { translation: Qt.vector3d( 7.5, -1.0, -10) } ] }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
                // 자차
                Entity {
                    components: [
                        SceneLoader {
                            id: carModel_my
                            source: "qrc:/icons/test.obj"
                            onStatusChanged: {
                                if (status === SceneLoader.Ready) console.log("자차
모델 로드 완료")
                                else if (status === SceneLoader.Error)
console.error("자차 모델 로드 실패")
                            }
                        },
                        Transform {
                            translation: Qt.vector3d(0, -0.9, 13)
                            scale3D: Qt.vector3d(0.03, 0.03, 0.03)
                            rotation: fromAxisAndAngle(Qt.vector3d(0, 1, 0), 90)
                        }
                    ]
                }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
                // 주변 차량들 (처음에 로드해두고 꺼둠: enabled=false)
                Entity {
                    id: carEnt1
                    enabled: true
                    components: [
                        SceneLoader { id: carModel1; source: "qrc:/icons/test.obj" },
                        Transform { translation: Qt.vector3d(-4.8, -1, -20); scale3D:
Qt.vector3d(0.03,0.03,0.03); rotation: fromAxisAndAngle(Qt.vector3d(0,1,0), -90) }
                    ]
                }
                Entity {
                    id: carEnt2
                    enabled: true
                    components: [
                        SceneLoader { id: carModel2; source: "qrc:/icons/test.obj" },
                        Transform { translation: Qt.vector3d(-4.8, -1,   2); scale3D:
Qt.vector3d(0.03,0.03,0.03); rotation: fromAxisAndAngle(Qt.vector3d(0,1,0), -90) }
                    ]
                }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
                Entity {
                    id: carEnt3
                    enabled: true
                    components: [
                        SceneLoader { id: carModel3; source: "qrc:/icons/test.obj" },
                        Transform { id: carTransform_1; translation: Qt.vector3d( 4.8, -1,   2);
scale3D: Qt.vector3d(0.03,0.03,0.03); rotation: fromAxisAndAngle(Qt.vector3d(0,1,0),  90) }
                    ]
                }
                Entity {
                    id: carEnt4
                    enabled: true
                    components: [
                        SceneLoader { id: carModel4; source: "qrc:/icons/test.obj" },
                        Transform { id: carTransform_2; translation: Qt.vector3d( 4.8, -1, -20);
scale3D: Qt.vector3d(0.03,0.03,0.03); rotation: fromAxisAndAngle(Qt.vector3d(0,1,0),  90) }
                    ]
                }
```

Panel App 구현

- qml.qrc/main.qml 코드 수정

```
                Entity {
                    id: carEnt5
                    enabled: true
                    components: [
                        SceneLoader { id: carModel5; source: "qrc:/icons/test.obj" },
                        Transform { translation: Qt.vector3d(0, -0.9, -5); scale3D:
Qt.vector3d(0.03,0.03,0.03); rotation: fromAxisAndAngle(Qt.vector3d(0,1,0),  90) }
                    ]
                }

                // 1초 뒤 끄기 (enabled=false). 더 이상 source="" 하지 않음!
                Timer {
                    interval: 200
                    repeat: true
                    running: true
                    onTriggered: {
                        var now = Date.now()
                        if (now - lastSeenMs[1] > 1000) carEnt1.enabled = false
                        if (now - lastSeenMs[2] > 1000) carEnt2.enabled = false
                        if (now - lastSeenMs[3] > 1000) carEnt3.enabled = false
                        if (now - lastSeenMs[4] > 1000) carEnt4.enabled = false
                        if (now - lastSeenMs[5] > 1000) carEnt5.enabled = false
```

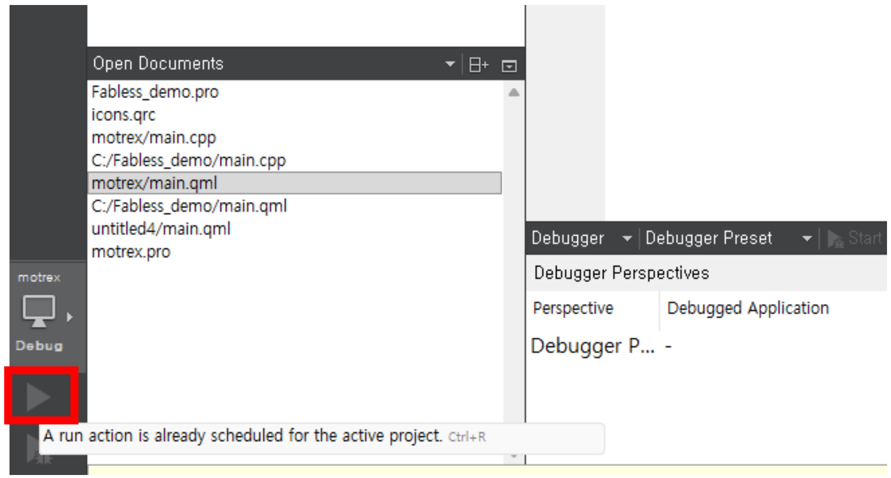
Panel App 구현

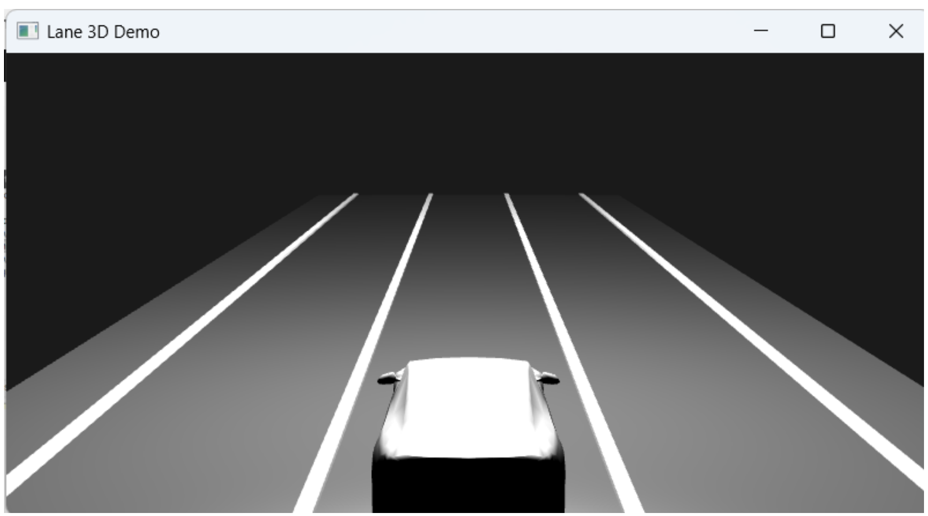
- qml.qrc/main.qml 코드 수정

```
                    }
                }
            }
        }
    }

    Component.onCompleted: updateCameraPosition()
}
```


Telechips

Panel App 구현





- 작성이 완료되면 run 실행을 통해 검증
- 다음과 같은 qt 화면 확인

Panel App 구현 확인

```
int main(int argc, char *argv[])
{
    QGuiApplication app(argc, argv);

    //QSurfaceFormat format;
    //format.setRenderableType(QSurfaceFormat::OpenGLES);
    //format.setVersion(2, 0);
    //QSurfaceFormat::setDefaultFormat(format);

    // --- 커맨드라인 파서 설정 ---
    QCommandLineParser parser;
    parser.setApplicationDescription("Lane telemetry client");
    parser.addHelpOption();
    QCommandLineOption ipOpt({"i","ip"}, "Server IP address", "ip");
    QCommandLineOption portOpt({"p","port"}, "Server port", "port");
```

```
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    parser.addOption(ipOpt);
```
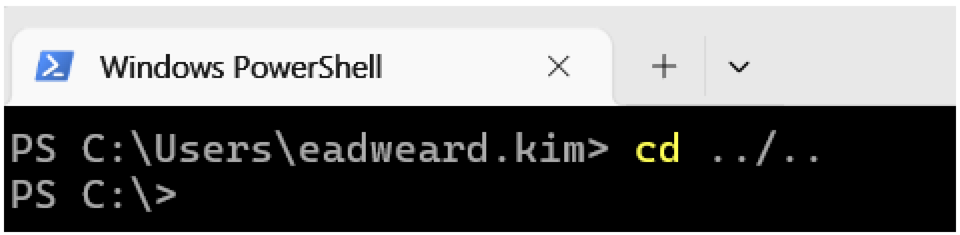

- main.cpp에 주석 해제

Panel App 구현 확인

- QT panel 보드로 전송(Host PC와 Board가 같은 wifi여야함)
  - D3-G보드에서 ifconfig 명령어로 ip(wlan0) 확인
  - Power shell 실행
  - cd .. 명령어를 통해 c 드라이브로 이동

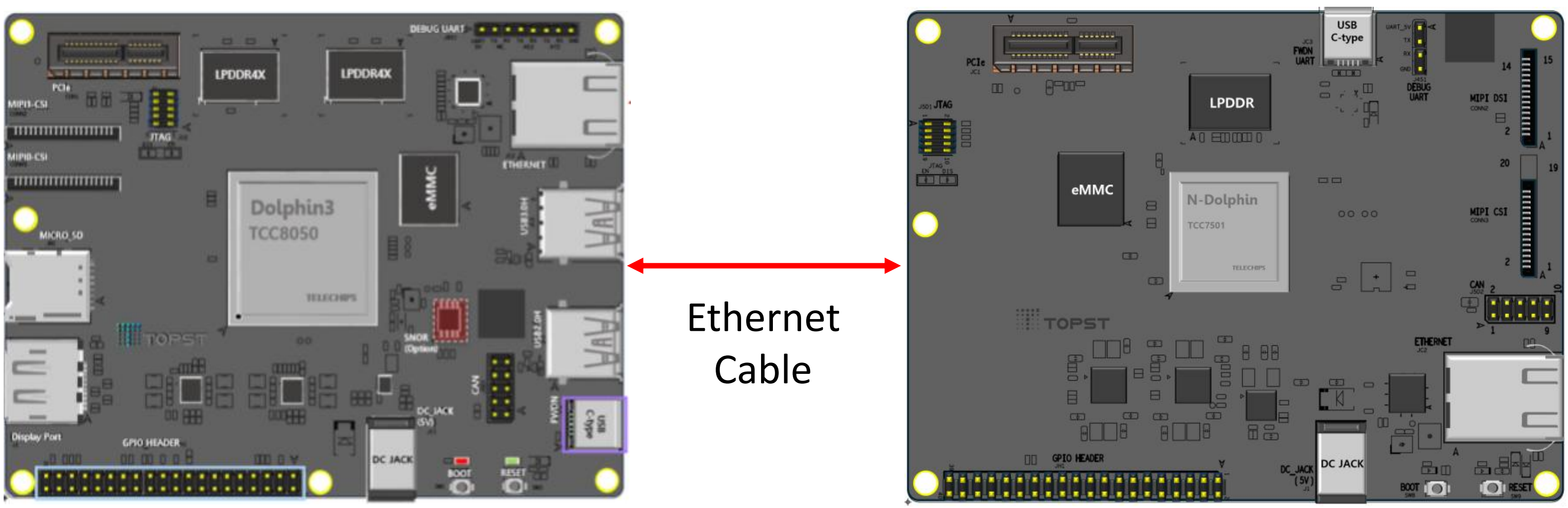


  - $ scp -r Fabless_demo topst@{확인한 ip}:/home/topst
  - 암호 topst입력

Sensing Zone + HPC Zone

Panel App 구현 확인

- D3-G 보드와 AI-G를 랜선으로 연결

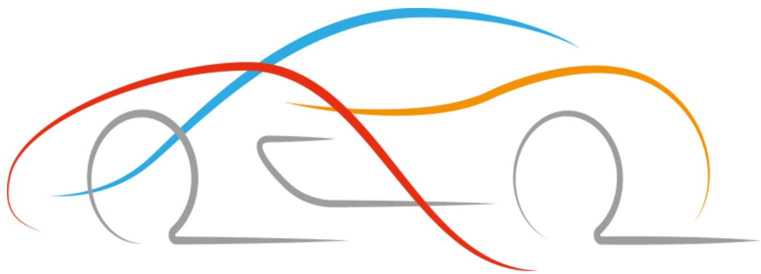


Panel App 구현 확인

- 해당 강의자료에서는 사전 빌드된 앱 사용
- D3-G 보드의 IP 설정
  - $ sudo ifconfig eth0 192.168.0.101 up
- Fabless앱 디렉터리로 이동
  - $ cd Fabless_Education_blink/Step3/fabless
  - $ sudo chmod 755 *
  - $ sudo make clean
  - $ sudo qmake
  - $ sudo make
  - $ ./fabless -i 192.168.0.100 -p 9999
  - AI-G 추론 시작 및 D3-G 패널 앱 정상 동작 확인

Panel App 구현 확인

- 만약 D3-G보드의 wlan0의 ip가 192.168.0.xxx라서 eth0와 겹칠 경우
- AI-G 보드 IP 수정 -> sudo ifconfig eth0 192.168.10.100 up
- D3-G 보드 IP 수정 -> sudo ifconfig eth0 192.168.10.101 up
- D3-G 패널 앱 실행 명령어 -> ./fabless -i 192.168.10.100 -p 9999



Thank You